EE4047 – High Voltage Techniques I

New Topic:
Generation of High Voltage Engineerings

Week 4.1 - Monday Lesson

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2.2 Electrostatic generators

Electrostatic generators convert mechanical energy directly into electrical energy. In contrast to elegromagnetic energy conversion, electrical charges are moved in this generator against the force of electrical fields, gaining higher potential energies and consuming mechanical energy.

Charge is sprayed onto an insulating moving belt by means of corona discharge points or direct contact. The belt, is driven by means of a motor and the charge is conveyed to the upper end where it is removed from the belt by discharging points connected to the inside of an insulated metal electrode.

The potential of the HV terminal at any instant is:
$V = Q/C$

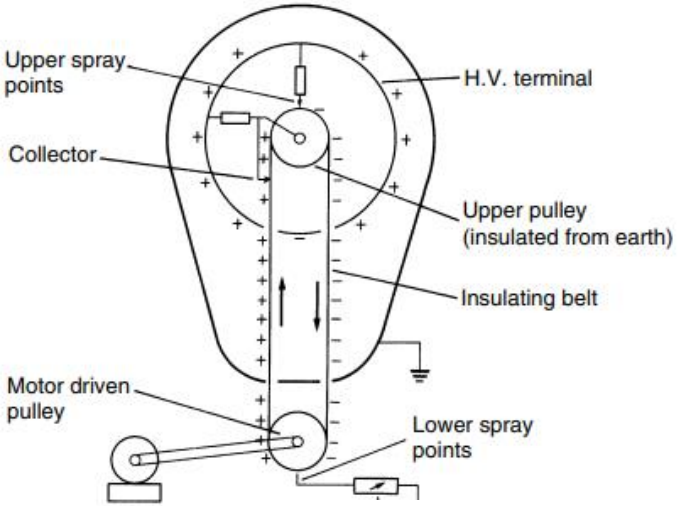


Figure 2.8 *Outli*

3 ALTERNATING VOLTAGES

Although power transmission systems are mostly of three-phase type, the testing voltages are usually single-phase voltages to ground. The rms value is for a cycle of T

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T V^2(t)dt}$$

Power dissipation formula:

$$P_n = kV_n{}^2\omega C_t$$

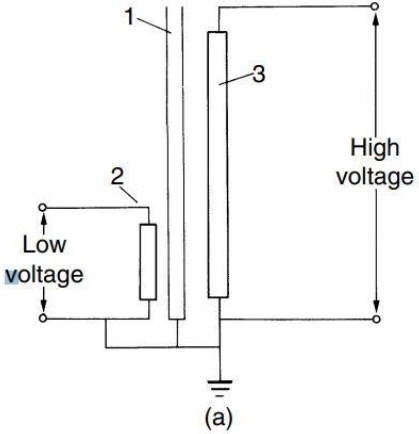
in which the factor $k \geq 1$ is used for additional capacitances withing the whole test circuit and some safety factor.

Nominal currents can be calcuated using equation:

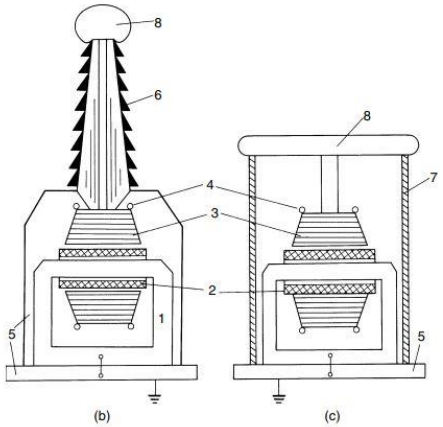
$$I_n = {}^{P_n}/_{V_n}$$

3.1 Testing Transformers

The power frequency single-phase transformer is the most common form of HV testing apparatus. The main differences compared to typical power transformer are smaller flux density withing the core to avoid unnecessary high magnetizing currents which would produce higher harmonics in the voltage regulator supplying the transformer, and a very compact and well insulated HV winding for the rated voltage.



(a)



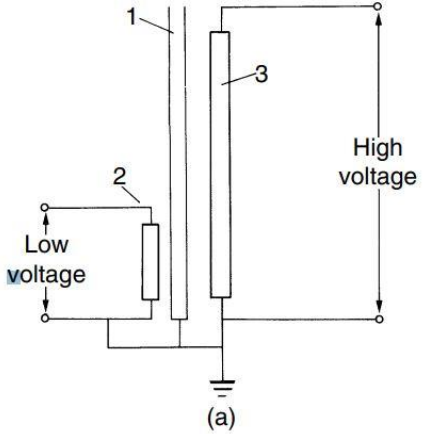
(b) (c)

Figure 2.12 *Single unit testing transformers. (a) Diagram. (b & c) different construction units. (1) Iron core. (2) Primary l.v. or exciting winding. (3) Secondary h.v. winding. (4) Field grading shield. (5) Grounded metal tank and base. (6) H.V. bushing. (7) Insulating shell or tank. (8) H.V. electrode*

3 ALTERNATING VOLTAGES

The primary winding ‘2’ is usually rated for low voltages of $\leq 1kV$, but might often be split up in two or more windings which can be switched in series or parallel to increase the regulation capabilities. The iron core ‘1’ is fixed at earth potential as well as one terminal of each of the two windings. In subfigure (b), the metal tank is grounded and a HV bushing ‘6’ is necessary to bring the HV out of the tank ‘5’. Instead of a bushing, a coaxial cable could also be used. In subfigure (c), the active part of the transformer is housed within an isolating cylinder ‘7’ avoiding the use of the bushing. This construction reduces the height.

The vessels are filled with high-quality transformer oil, as most of the windings are oil-paper insulated.



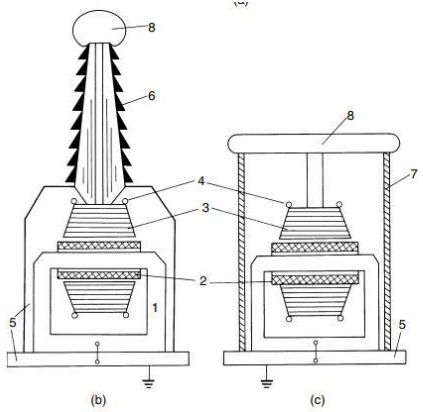


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3 ALTERNATING VOLTAGES

Design of the HV winding becomes difficullt if voltages of more than some 100 kV must be produced within one coil. Better constructions are available by specialized techniques, mainly by cascading transfomers.

Cascadded transformers

For voltages higher than about 300 to 500 kV, the cascading of transformers is a big advantage, as the whole testing requirements can be subdivided into single units and therefore transport and setup becomes easier.

The LV supply is connected to the primary winding ‘1’ of transformer I, designed for an HV output of V as are the other two trafos.

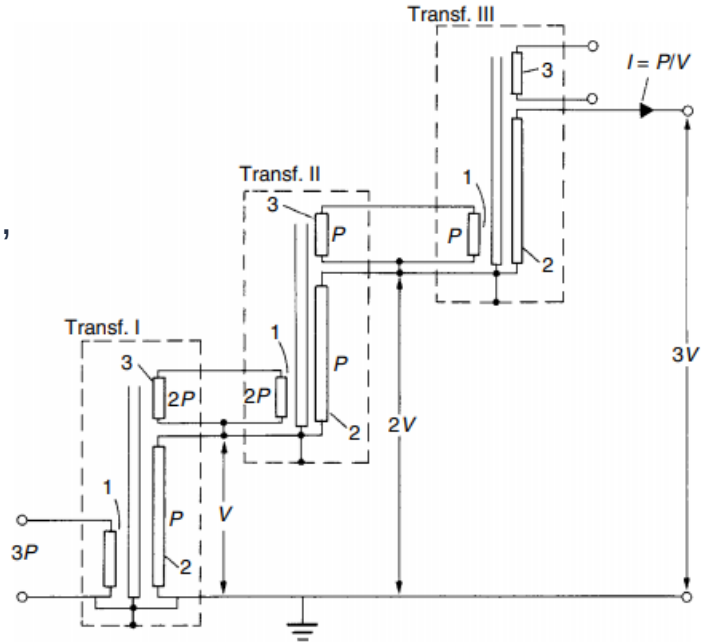


Figure 2.15 *Basic circuit of cascaded transformers. (1) Primary windings. (2) Secondary h.t. windings. (3) Tertiary exciting windings*

3 ALTERNATING VOLTAGES

The winding ‘3’ supplies the primary of the second transformer unit II, both windings are dimensioned for the same low voltage and the potential is fixed to the high potential V. The secondary winings ‘2’ of both units are series connected so that a voltage of 2 V is produced hereby. The addition of the stage III is the same. The tanks or vessels containing the active parts (core and windings) are indicated by dashed lines only.

In the figure P shows the loading of primary windings, the product of current and voltage for each of the coils. The primary windings for the lower stages are heavily loaded.

Each of the windings ‘2’ will carry a current of $I = P/V$

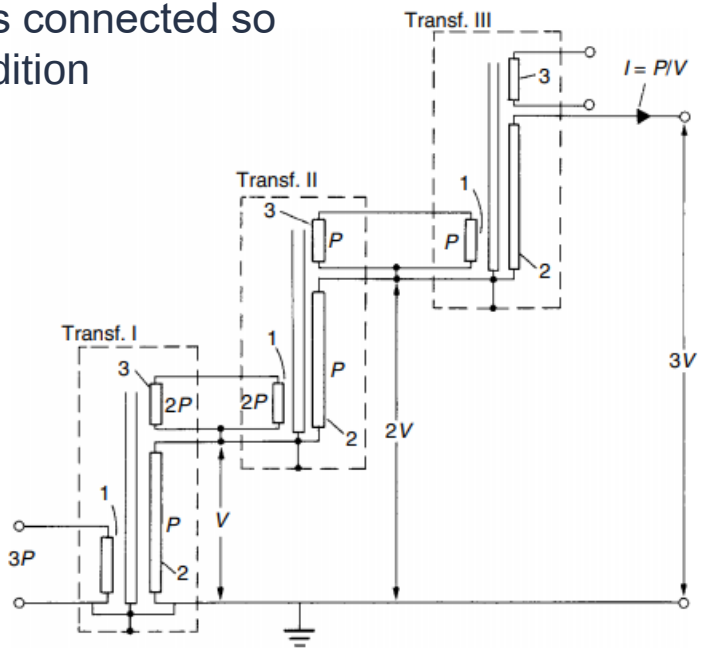


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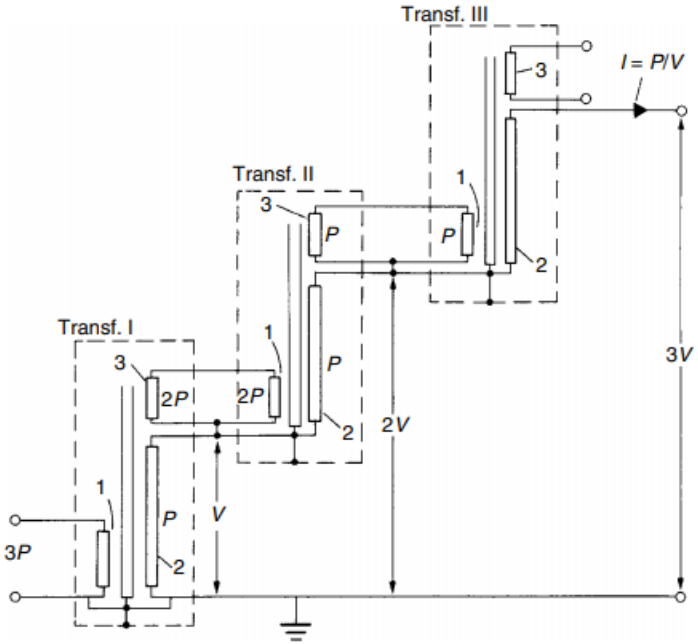


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3 ALTERNATING VOLTAGES

The leakge or stray reactances X for each winding are the primary X_p, the secondary HV winding X_h and the exciting winding X_e. Neglecting losses within the windings and magnetizing currents, the simplified calculation of the resultant ractance of a cascade unit with n transformers:

$$X_{res} = \sum_{v=1}^{n} \left[X_h + (n-v)^2 X_e + (n+1-v)^2 X_p \right]$$

Assuming three equal transformer units, the equation leads to a resultant reactance of

$$X_{res} = 3X_h + 5X_e + 14X_p$$

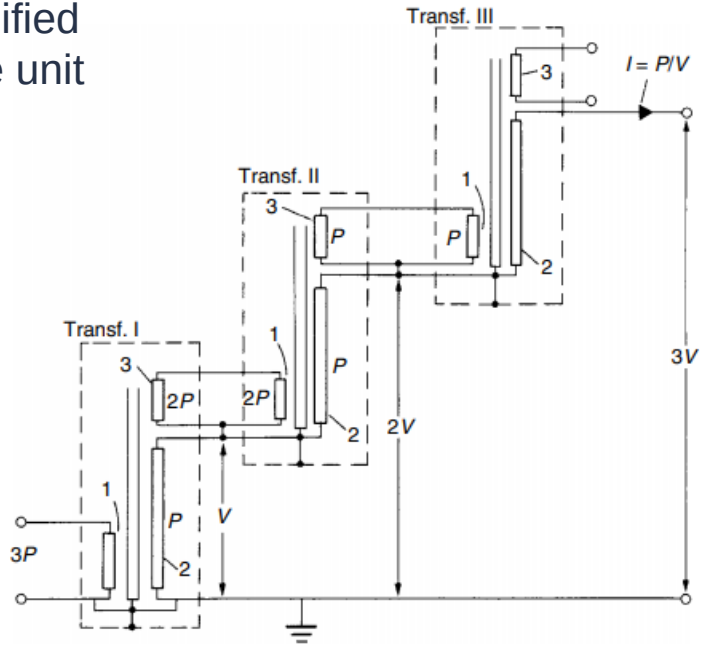


Figure 2.15 *Basic circuit of cascaded transformers. (1) Primary windings. (2) Secondary h.t. windings. (3) Tertiary exciting windings*

END OF THE LESSON

Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm